

1 Specify the diagnostic target

Marginal

Gene-wise or aggregated KS

Covariate-indexed

Residual variation over pseudotime, batch, state

Dependence

Rank correlation or empirical copula

2 Fit the model

$$\hat{\theta} = \mathcal{A}(D)$$

Conditional count model

3 Randomized PIT

$$U_i = F_{\hat{\theta}}(X_i^- | Z_i) + V_i p_{\hat{\theta}}(X_i | Z_i)$$

4 Compute statistic

$$S_{\text{obs}} = S(U, D)$$

Target-specific statistic

$V_i \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$, independent of the observed data

5 Refitted bootstrap: repeat B times

Simulate

$$D_b^* \sim P_{\hat{\theta}}$$

Preserve the design

Refit

$$\hat{\theta}_b^* = \mathcal{A}(D_b^*)$$

Same fitting algorithm

Recompute

Fresh independent PIT and the same statistic S_b^*

Compare S_{obs} with the empirical distribution of $S_1^*, \dots, S_B^*$

6 Draw a component-specific conclusion

$$\hat{p} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{S_b^* \geq S_{\text{obs}}\}}{B + 1}$$

Apply the prespecified multiplicity rule; report only the tested component.